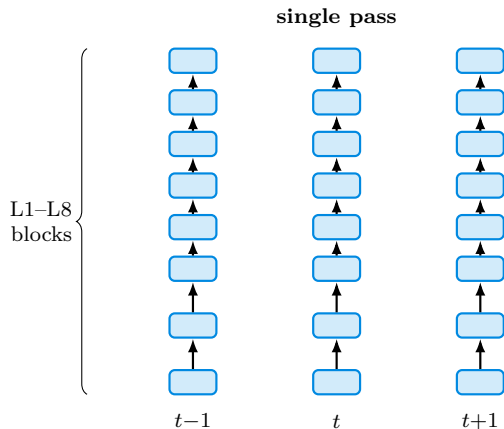


Training-time information flows

Depth recurrence OFF

Vanilla transformer

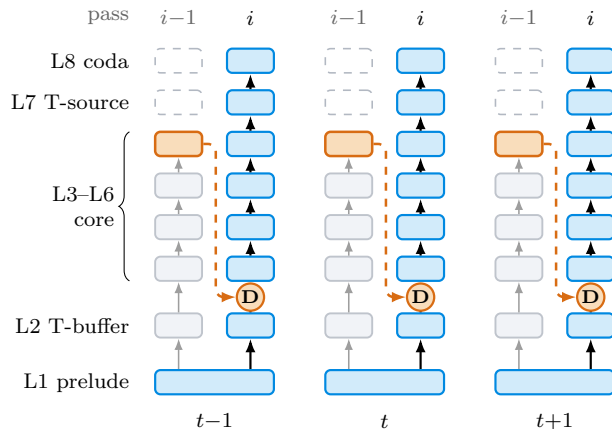
One pass through L1–L8 at every position
(embedding and head not shown).



Depth recurrence ON

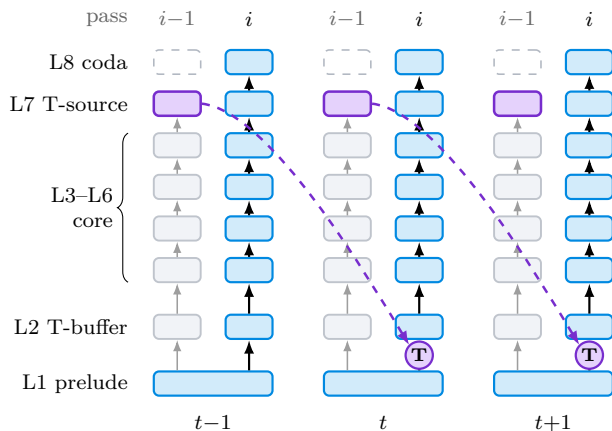
Depth-recurrent / looped LM

Pass $i-1$ writes the core output (L6) as depth state;
pass i reads it at the same position, just before the core.



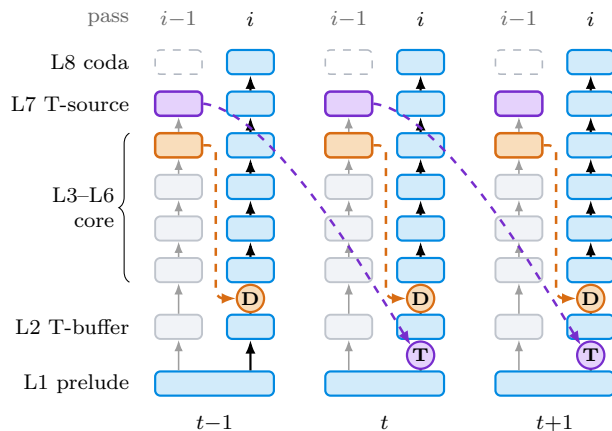
Temporally recurrent

Pass $i-1$ writes the T-source output (L7) as temporal state;
pass i reads it one position later, just after the prelude.



Depth + temporal hybrid

Both at once: depth state stays at its position (short orange arrows),
temporal state moves one position forward (long purple arrows).



pass $i-1$ layer

pass i layer

not run on this pass

prelude: runs once

writes depth state

writes temporal state

depth read (same position)

temporal read (next position)